Ansible Automation - Use Cases

A reference of the use cases by which Ansible Automation Platform is commonly implemented, by domain. Each use case is described in the same format (Description, Collection, Key content, Notes) and mapped to Red Hat Certified and Validated content.

4 solution use cases · 13 product use cases · generated 2026-10-21 · community reference, not an official Red Hat publication

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Google Cloud

Provision and operate Google Cloud resources.

Solution use cases

Use Case: Provision Google Cloud networking and compute

Description	Build a Google Cloud environment with a playbook: a VPC network, subnets, firewall rules, IP addresses and Compute Engine instances. Projects can be rebuilt consistently in development, test and production, and removed when finished.
Collection	`google.cloud` (Certified, v1.15.0, updated 2026-10-01), `cloud.gcp_ops` (Validated, v2.0.0, updated 2024-08-01)
Key content	• `google.cloud.gcp_compute_network` — Creates a GCP Network • `google.cloud.gcp_compute_subnetwork` — Creates a GCP Subnetwork • `google.cloud.gcp_compute_firewall` — Creates a GCP Firewall • `google.cloud.gcp_compute_address` — Creates a GCP Address • `google.cloud.gcp_compute_instance` — Creates a GCP Instance • `google.cloud.gcp_http_lb` — Create a HTTP Load Balancer on GCP • `cloud.gcp_ops.manage_machine_image` — cloud.gcp_ops.manage_machine_image
Notes	• Dynamic discovery of the resulting instances is covered in Use Google Cloud as a dynamic inventory source. • Resource creation order matters: create networks before subnets, and delete in reverse. • If Terraform is the standard provisioner, see Run Terraform from Automation Platform.

Use Case: Manage Google Cloud storage and services

Description	Define the supporting services of a Google Cloud project in code: storage buckets, managed databases, messaging topics, service accounts, encryption keys and secrets. Teams request services through a job template, and the configuration is the same across projects.
Collection	`google.cloud` (Certified, v1.15.0, updated 2026-10-01), `cloud.gcp_ops` (Validated, v2.0.0, updated 2024-08-01)
Key content	• `google.cloud.gcp_storage_bucket` — Creates a GCP Bucket • `google.cloud.gcp_sql_instance` — Creates a GCP Instance • `google.cloud.gcp_pubsub_topic` — Creates a GCP Topic • `google.cloud.gcp_iam_service_account` — Creates a GCP ServiceAccount • `google.cloud.gcp_kms_key_ring` — Creates a GCP KeyRing • `google.cloud.gcp_secret_manager` — Get Secrets from Google Cloud as a Lookup plugin • `cloud.gcp_ops.upload_file_to_cloud_storage.upload_file_to_cloud_storage` — Upload a file to GCP Cloud Storage
Notes	• Do not store service account keys in Git; prefer short-lived credentials or credentials held in Automation Platform. • The `google.cloud.gcp_secret_manager` lookup can fetch a secret at run time. • Related: Provision Google Cloud networking and compute.

Use Case: Use Google Cloud as a dynamic inventory source

Description	Use Google Cloud as the source of truth for which Compute Engine instances exist. Automation Platform syncs instances from one or more projects, and groups them by label, zone or machine type so jobs always target current machines.

Collection	`google.cloud` (Certified, v1.15.0, updated 2026-10-01)
Key content	• `google.cloud.gcp_compute` — Google Cloud Compute Engine inventory source • `google.cloud.gcp_compute_instance` — Creates a GCP Instance • `google.cloud.gcp_compute_instance_info` — Gather info for GCP Instance
Notes	• Pair with Event-Driven Ansible to run a job when a new instance appears, if an event source for the organization's messaging service is available. • Restrict the service account to read-only roles for inventory. • Instance details beyond inventory can be gathered with the `google.cloud.gcp_compute_instance_info` module.

Use Case: Migrate on-premises VMs to Google Cloud

Description	Move a virtual machine from an on-premises KVM hypervisor to Google Cloud with a validated playbook. The automation clones the VM, uploads the image to Cloud Storage, imports it as a Compute Engine image and starts an instance from it.
Collection	`cloud.gcp_ops` (Validated, v2.0.0, updated 2024-08-01), `google.cloud` (Certified, v1.15.0, updated 2026-10-01)
Key content	• `cloud.gcp_ops.move_vm_from_on_prem_to_gcp.move_vm_from_on_prem_to_gcp` — Playbook to migrate an existing on prem VM running on KVM hypervisor to GCP • `cloud.gcp_ops.clone_on_prem_vm` — clone_on_prem_vm • `cloud.gcp_ops.import_image_and_run_gcp_instance` — cloud.gcp_ops.import_image_and_run_gcp_instance • `cloud.gcp_ops.upload_file_to_cloud_storage.upload_file_to_cloud_storage` — Upload a file to GCP Cloud Storage • `google.cloud.gcp_compute_image` — Creates a GCP Image
Notes	• The validated playbook targets KVM sources. • The same pattern for AWS is in Migrate on-premises VMs to AWS. • Use `cloud.gcp_ops.manage_machine_image` to maintain reusable machine images after migration.

Product use cases

Grouped from the modules of each certified or validated collection. Key content lists the components each use case supports, with the modules that manage them (create, update, delete) and query them (_info).

Google Cloud Platform `cloud.gcp_ops`

Validated Red Hat · version 2.0.0 · updated 2024-08-01 · 0 modules → 1 use cases

Use Case: Migrate VMs and manage images on Google Cloud

Description	Clone on-premises KVM VMs, import images and run GCP instances, manage machine images and upload files to Cloud Storage.
Collection	`cloud.gcp_ops` (Validated, Red Hat, v2.0.0, updated 2024-08-01)
Key content	**VM migration playbook** playbook `move_vm_from_on_prem_to_gcp.move_vm_from_on_prem_to_gcp` **Clone on-premises VM** role `clone_on_prem_vm` **Import image and run instance** role `import_image_and_run_gcp_instance` **Machine images** role `manage_machine_image` **Cloud Storage upload playbook** playbook `upload_file_to_cloud_storage.upload_file_to_cloud_storage`

Google Cloud Platform `google.cloud`

Certified Google Cloud · version 1.15.0 · updated 2026-10-01 · 225 modules → 12 use cases

Use Case: Manage Compute Engine instances and fleets

Description	Create VM instances from templates and images, group and autoscale them, reserve capacity, use sole-tenant nodes and TPUs, and build inventory or reach hosts through IAP.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)
Key content	**VM instances** `gcp_compute_instance`, `gcp_compute_instance_info` **Instance templates** `gcp_compute_instance_template`, `gcp_compute_instance_template_info` **Images** `gcp_compute_image`, `gcp_compute_image_info` **Instance groups** `gcp_compute_instance_group`, `gcp_compute_instance_group_info` **Managed instance groups** `gcp_compute_instance_group_manager`, `gcp_compute_instance_group_manager_info` **Regional managed instance groups** `gcp_compute_region_instance_group_manager`, `gcp_compute_region_instance_group_manager_info` **Autoscalers** `gcp_compute_autoscaler`, `gcp_compute_autoscaler_info` **Regional autoscalers** `gcp_compute_region_autoscaler`, `gcp_compute_region_autoscaler_info` **Reservations** `gcp_compute_reservation`, `gcp_compute_reservation_info` **Resource policies** `gcp_compute_resource_policy`, `gcp_compute_resource_policy_info` **Node templates** `gcp_compute_node_template`, `gcp_compute_node_template_info` **Node groups** `gcp_compute_node_group`, `gcp_compute_node_group_info` **TPU nodes** `gcp_tpu_node`, `gcp_tpu_node_info` **Compute Engine inventory** inventory `gcp_compute` **SSH through Identity-Aware Proxy** connection `iap` **Google Cloud SDK installation** role `gcloud` **gcsfuse installation** role `gcsfuse`
Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Manage disks, file shares and object storage

Description	Create persistent disks and snapshots, Filestore instances, Cloud Storage buckets with access controls, and upload or download objects.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)
Key content	**Disks** `gcp_compute_disk`, `gcp_compute_disk_info` **Regional disks** `gcp_compute_region_disk`, `gcp_compute_region_disk_info` **Snapshots** `gcp_compute_snapshot`, `gcp_compute_snapshot_info` **Filestore instances** `gcp_filestore_instance`, `gcp_filestore_instance_info` **Storage buckets** `gcp_storage_bucket` **Bucket access controls** `gcp_storage_bucket_access_control` **Default object ACLs** `gcp_storage_default_object_acl` **Storage objects** `gcp_storage_object`
Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Manage VPC networks, firewalls and DNS

Description	Create VPC networks, subnetworks, routes, Cloud Routers, static addresses, network endpoint groups, firewall rules and Cloud DNS zones and records.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)

Key content	**VPC networks** `gcp_compute_network`, `gcp_compute_network_info` **Subnetworks** `gcp_compute_subnetwork`, `gcp_compute_subnetwork_info` **Routes** `gcp_compute_route`, `gcp_compute_route_info` **Cloud Routers** `gcp_compute_router`, `gcp_compute_router_info` **Addresses** `gcp_compute_address`, `gcp_compute_address_info` **Global addresses** `gcp_compute_global_address`, `gcp_compute_global_address_info` **Network endpoint groups** `gcp_compute_network_endpoint_group`, `gcp_compute_network_endpoint_group_info` **Firewall rules** `gcp_compute_firewall`, `gcp_compute_firewall_info` **App Engine firewall rules** `gcp_appengine_firewall_rule`, `gcp_appengine_firewall_rule_info` **DNS managed zones** `gcp_dns_managed_zone`, `gcp_dns_managed_zone_info` **DNS record sets** `gcp_dns_resource_record_set`, `gcp_dns_resource_record_set_info`
Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Connect to on-premises networks

Description	Set up Cloud VPN gateways and tunnels and Interconnect attachments.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)
Key content	**External VPN gateways** `gcp_compute_external_vpn_gateway`, `gcp_compute_external_vpn_gateway_info` **Target VPN gateways** `gcp_compute_target_vpn_gateway`, `gcp_compute_target_vpn_gateway_info` **VPN tunnels** `gcp_compute_vpn_tunnel`, `gcp_compute_vpn_tunnel_info` **Interconnect attachments** `gcp_compute_interconnect_attachment`, `gcp_compute_interconnect_attachment_info`
Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Configure load balancing

Description	Build HTTP(S), SSL, TCP and network load balancers from backend services, health checks, forwarding rules, proxies, URL maps and certificates.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)

Key content	**Backend services** `gcp_compute_backend_service`, `gcp_compute_backend_service_info` **Regional backend services** `gcp_compute_region_backend_service`, `gcp_compute_region_backend_service_info` **Backend buckets** `gcp_compute_backend_bucket`, `gcp_compute_backend_bucket_info` **Health checks** `gcp_compute_health_check`, `gcp_compute_health_check_info` **Regional health checks** `gcp_compute_region_health_check`, `gcp_compute_region_health_check_info` **HTTP health checks** `gcp_compute_http_health_check`, `gcp_compute_http_health_check_info` **HTTPS health checks** `gcp_compute_https_health_check`, `gcp_compute_https_health_check_info` **Forwarding rules** `gcp_compute_forwarding_rule`, `gcp_compute_forwarding_rule_info` **Global forwarding rules** `gcp_compute_global_forwarding_rule`, `gcp_compute_global_forwarding_rule_info` **Target HTTP proxies** `gcp_compute_target_http_proxy`, `gcp_compute_target_http_proxy_info` **Target HTTPS proxies** `gcp_compute_target_https_proxy`, `gcp_compute_target_https_proxy_info` **Regional target HTTP proxies** `gcp_compute_region_target_http_proxy`, `gcp_compute_region_target_http_proxy_info` **Regional target HTTPS proxies** `gcp_compute_region_target_https_proxy`, `gcp_compute_region_target_https_proxy_info` **Target SSL proxies** `gcp_compute_target_ssl_proxy`, `gcp_compute_target_ssl_proxy_info` **Target TCP proxies** `gcp_compute_target_tcp_proxy`, `gcp_compute_target_tcp_proxy_info` **Target pools** `gcp_compute_target_pool`, `gcp_compute_target_pool_info` **Target instances** `gcp_compute_target_instance`, `gcp_compute_target_instance_info` **URL maps** `gcp_compute_url_map`, `gcp_compute_url_map_info` **Regional URL maps** `gcp_compute_region_url_map`, `gcp_compute_region_url_map_info` **SSL certificates** `gcp_compute_ssl_certificate`, `gcp_compute_ssl_certificate_info` **SSL policies** `gcp_compute_ssl_policy`, `gcp_compute_ssl_policy_info` **HTTP load balancer role** role `gcp_http_lb`
Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Run Kubernetes clusters

Description	Create Google Kubernetes Engine clusters and node pools.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)
Key content	**GKE clusters** `gcp_container_cluster`, `gcp_container_cluster_info` **Node pools** `gcp_container_node_pool`, `gcp_container_node_pool_info`
Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Build and secure container delivery

Description	Define Cloud Build triggers and repository connections, Binary Authorization attestors and policies, and Container Analysis notes.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)
Key content	**Cloud Build triggers** `gcp_cloudbuild_trigger`, `gcp_cloudbuild_trigger_info` **Cloud Build connections** `gcp_cloudbuildv2_connection`, `gcp_cloudbuildv2_connection_info` **Cloud Build repositories** `gcp_cloudbuildv2_repository`, `gcp_cloudbuildv2_repository_info` **Binary Authorization attestors** `gcp_binaryauthorization_attestor`, `gcp_binaryauthorization_attestor_info` **Binary Authorization policies** `gcp_binaryauthorization_policy`, `gcp_binaryauthorization_policy_info` **Container Analysis notes** `gcp_containeranalysis_note`, `gcp_containeranalysis_note_info`

Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Run serverless functions and messaging

Description	Deploy Cloud Functions, schedule jobs, queue tasks and manage Pub/Sub topics and subscriptions.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)
Key content	**Cloud Functions** `gcp_cloudfunctions_cloud_function`, `gcp_cloudfunctions_cloud_function_info` **Scheduler jobs** `gcp_cloudscheduler_job`, `gcp_cloudscheduler_job_info` **Task queues** `gcp_cloudtasks_queue`, `gcp_cloudtasks_queue_info` **Pub/Sub topics** `gcp_pubsub_topic`, `gcp_pubsub_topic_info` **Pub/Sub subscriptions** `gcp_pubsub_subscription`, `gcp_pubsub_subscription_info`
Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Manage databases, caches and analytics

Description	Create Cloud SQL, AlloyDB, Spanner, Bigtable and Memorystore for Redis instances, their databases, users and backups, and BigQuery datasets and tables.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)
Key content	**Cloud SQL instances** `gcp_sql_instance`, `gcp_sql_instance_info` **Cloud SQL databases** `gcp_sql_database`, `gcp_sql_database_info` **Cloud SQL users** `gcp_sql_user`, `gcp_sql_user_info` **Cloud SQL SSL certificates** `gcp_sql_ssl_cert` **AlloyDB clusters** `gcp_alloydb_cluster`, `gcp_alloydb_cluster_info` **AlloyDB instances** `gcp_alloydb_instance`, `gcp_alloydb_instance_info` **AlloyDB users** `gcp_alloydb_user`, `gcp_alloydb_user_info` **AlloyDB backups** `gcp_alloydb_backup`, `gcp_alloydb_backup_info` **Spanner instances** `gcp_spanner_instance`, `gcp_spanner_instance_info` **Spanner databases** `gcp_spanner_database`, `gcp_spanner_database_info` **Bigtable instances** `gcp_bigtable_instance`, `gcp_bigtable_instance_info` **Redis instances** `gcp_redis_instance`, `gcp_redis_instance_info` **BigQuery datasets** `gcp_bigquery_dataset`, `gcp_bigquery_dataset_info` **BigQuery tables** `gcp_bigquery_table`, `gcp_bigquery_table_info`
Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Manage projects, service accounts and APIs

Description	Create projects, enable APIs, define IAM roles and service accounts with keys, and create log-based metrics.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)
Key content	**Projects** `gcp_resourcemanager_project`, `gcp_resourcemanager_project_info` **Enabled APIs** `gcp_serviceusage_service`, `gcp_serviceusage_service_info` **IAM roles** `gcp_iam_role`, `gcp_iam_role_info` **Service accounts** `gcp_iam_service_account`, `gcp_iam_service_account_info` **Service account keys** `gcp_iam_service_account_key` **Log-based metrics** `gcp_logging_metric`, `gcp_logging_metric_info`

Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Protect secrets, parameters and encryption keys

Description	Create KMS key rings and keys, read and update Secret Manager secrets and Parameter Manager parameters, store runtime configuration, and retrieve them in playbooks.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)
Key content	**KMS key rings** `gcp_kms_key_ring`, `gcp_kms_key_ring_info` **KMS crypto keys** `gcp_kms_crypto_key`, `gcp_kms_crypto_key_info` **Secret Manager secrets** `gcp_secret_manager` **Parameter Manager parameters** `gcp_parameter_manager` **Runtime configs** `gcp_runtimeconfig_config`, `gcp_runtimeconfig_config_info` **Runtime config variables** `gcp_runtimeconfig_variable`, `gcp_runtimeconfig_variable_info` **Secret Manager lookup** lookup `gcp_secret_manager` **Parameter Manager lookup** lookup `gcp_parameter_manager` **KMS auth token filter** filter `gcp_kms_filters`
Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.

Use Case: Build machine learning on Vertex AI and Colab

Description	Manage Vertex AI datasets, endpoints, feature stores, vector indexes, metadata, TensorBoard and reasoning engines, and Colab runtimes, schedules and notebook executions.
Collection	`google.cloud` (Certified, Google Cloud, v1.15.0, updated 2026-10-01)
Key content	**Datasets** `gcp_vertexai_dataset`, `gcp_vertexai_dataset_info` **Endpoints** `gcp_vertexai_endpoint`, `gcp_vertexai_endpoint_info` **Model Garden endpoints** `gcp_vertexai_endpoint_with_model_garden_deployment` **Deployment resource pools** `gcp_vertexai_deployment_resource_pool`, `gcp_vertexai_deployment_resource_pool_info` **Feature stores** `gcp_vertexai_featurestore`, `gcp_vertexai_featurestore_info` **Feature store entity types** `gcp_vertexai_featurestore_entitytype`, `gcp_vertexai_featurestore_entitytype_info` **Feature store entity type features** `gcp_vertexai_featurestore_entitytype_feature`, `gcp_vertexai_featurestore_entitytype_feature_info` **Feature groups** `gcp_vertexai_feature_group`, `gcp_vertexai_feature_group_info` **Feature group features** `gcp_vertexai_feature_group_feature`, `gcp_vertexai_feature_group_feature_info` **Online feature stores** `gcp_vertexai_feature_online_store`, `gcp_vertexai_feature_online_store_info` **Online feature views** `gcp_vertexai_feature_online_store_featureview`, `gcp_vertexai_feature_online_store_featureview_info` **Vector indexes** `gcp_vertexai_index`, `gcp_vertexai_index_info` **Index endpoints** `gcp_vertexai_index_endpoint`, `gcp_vertexai_index_endpoint_info` **Deployed indexes** `gcp_vertexai_index_endpoint_deployed_index` **Metadata stores** `gcp_vertexai_metadata_store`, `gcp_vertexai_metadata_store_info` **RAG engine configuration** `gcp_vertexai_rag_engine_config` **Reasoning engines** `gcp_vertexai_reasoning_engine`, `gcp_vertexai_reasoning_engine_info` **TensorBoards** `gcp_vertexai_tensorboard`, `gcp_vertexai_tensorboard_info` **Colab runtimes** `gcp_colab_runtime`, `gcp_colab_runtime_info` **Colab runtime templates** `gcp_colab_runtime_template`, `gcp_colab_runtime_template_info` **Colab schedules** `gcp_colab_schedule`, `gcp_colab_schedule_info` **Colab notebook executions** `gcp_colab_notebook_execution`, `gcp_colab_notebook_execution_info`
Notes	• Most resource modules cover both create/update/delete and read-only info queries, and are named gcp_<resource>.